

Appendix

for “Disaster-Exposed Peers and Unequal Academic Loss: Evidence from the Wenchuan Earthquake

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A Classroom Assignment and Balance Evidence

For the coefficient on peer exposure to admit a causal interpretation, classroom peer composition must not be correlated with unobserved student characteristics in ways that would confound identification. The main concern is that a performance-based assignment rule could generate classroom-level ability clustering, in which case the estimated peer effect would partly reflect sorting on academic ability rather than spillovers from classmates' earthquake exposure.

School interviews indicate that classroom assignment followed a rank-based allocation rule that was tied to entrance-exam rankings but still dispersed students across classrooms rather than sorting them into high- and low-performing classes. In the regular track, students were not assigned by placing all top-ranked students in one class and all lower-ranked students in another. Instead, schools typically rotated students across classrooms in rank order: for example, rank 1 was placed in Class 1, rank 2 in Class 2, rank 3 in Class 3, and rank 4 returned to Class 1, with the same pattern repeated down the ranking. This rule spreads students of different achievement levels across regular classrooms and therefore does not mechanically generate ability clustering across those classes. In some schools, the highest-ranked students were first separated into an advanced track; the remaining students were then allocated across regular classrooms using the same rank-based rule. The identifying assumption is therefore that, within school-grade cells, classroom allocation is plausibly as good as random with respect to unobserved determinants of later outcomes.

To assess this implication empirically, I conduct pairwise balance tests in the spirit of [Guo and Qu \(2022\)](#). Within each school-grade cell, I form feasible classroom pairs among classrooms in the same track, which yields 21 classroom pairs in total. Because the student's own old-school collapse status is the underlying treatment variable in this setting, I first examine whether that variable is balanced across classrooms. Neither the collapse indicator nor the raw five-point old-school damage scale yields any rejection at the 5% significance level in the pairwise t-tests or pairwise KS tests. This is a direct evidence that classrooms were not formed on earthquake exposure itself.

I then turn to other demographic characteristics: age, gender, Tibetan minority, parental education.¹ Among these and the old-school damage scale, 7.0% of the pairwise t-tests reject at the 5% significance level. This rate is close to the false-positive benchmark implied by repeated testing.² Combined with the treatment-variable results above, this evidence is consistent with an as-good-as-random allocation rule within school-grade cells rather than deliberate classroom

¹The two main non-Han groups in the survey area are Qiang and Tibetan. Tibetan minority is used as the ethnicity control in all regression specifications. For the rejection-rate calculation, I include the old-school damage scale but exclude the binary collapse indicator because it is mechanically derived from that scale.

²Across these five characteristics, Table J.3 reports 100 valid pairwise t-tests. Under exact balance, a 5% test level implies about 5.00 false rejections on average, or 5.0% of tests.

formation on earthquake exposure. Appendix Table J.3 and Appendix Figure K.4 report the full pairwise balance results.

Finally, own exposure is the individual-level component used to construct peer exposure. If it were related to predetermined student characteristics, peer exposure could partly capture classroom composition rather than the share of school-collapsed classmates. I therefore regress each measure of school damage on predetermined characteristics, conditional on the old school's distance from the earthquake epicenter. Appendix Table J.4 shows that none of these characteristics is individually significant at the 10% level. This evidence supports interpreting peer exposure as variation in classmates' earthquake exposure rather than observable pre-earthquake student composition.

B Robustness Checks

Appendix Figure K.5 presents a sensitivity analysis of the peer-exposure estimates in Table 1. It assesses whether the estimates persist when the treatment measure, sample coverage, or construction of baseline and outcome measures is modified. The first check replaces missing 2008 test scores with the subject-specific median while retaining the missing-score indicator. The second replaces the collapse dummy with a three-level exposure measure based on old-school damage severity. The third restricts the sample to students whose old school is observed in the survey data, thereby excluding the cases for which earthquake exposure is inferred from external records. The fourth restricts the sample to schools with greater within-school variation in exposure. The fifth addresses potential sample-selection concerns from missing 2010 administrative outcomes within the same 1,278-student full questionnaire sample. It replaces missing raw 2010 Chinese and Math scores with the corresponding self-reported raw scores when available, re-standardizes the combined outcomes, and re-estimates the fully controlled column (3) specification without adding school-grade observations outside the questionnaire sample.

The overall pattern again most clearly supports a negative peer effect for Chinese. The peer-exposure coefficient remains negative in all five checks. In the imputed-outcome exercise, combined Chinese and Math outcomes are available for 1,248 of the 1,278 questionnaire respondents, and the fully controlled estimation sample expands from 1,046 to 1,200 students. The peer-exposure coefficients are -4.46 for Chinese ($p = 0.03$), -3.12 for Math ($p = 0.08$), and -3.70 for Total score ($p = 0.03$). These estimates are close in magnitude to the corresponding administrative-only estimates of -4.43 , -3.31 , and -4.01 . Chinese and Total remain statistically significant at the 5% level, while the Math estimate is less precise but remains significant at the 10% level. Thus, within the same questionnaire sample and set of school-grade cells, the adverse peer effect is not an artifact of restricting the analysis to students

with non-missing administrative outcomes. Overall, the appendix robustness checks continue to show that the evidence for a negative peer effect is strongest for Chinese scores.

Moreover, classrooms with more school-collapsed peers may draw more heavily from a particular old school that collapsed. Peer exposure may therefore be correlated with observed classroom composition, including gender, ethnicity, and baseline academic scores, as well as with the shares of former schoolmates and classmates that proxy for pre-transition social networks. Appendix Table B.1 adds leave-one-out controls for these characteristics. Across all three outcomes, the peer-exposure coefficient remains negative and statistically significant and becomes more negative after these controls are included.³ Thus, the estimated adverse peer effect of school-collapsed classmates is not caused by these observed dimensions of classroom composition or by the measured composition of students' pre-transition social networks.

Table B.1: Robustness to additional classroom composition controls.

	Chinese		Math		Total	
	(1)	(2)	(1)	(2)	(1)	(2)
Own exposure	−0.33*** (0.01)	−0.31*** (0.01)	−0.38** (0.03)	−0.39** (0.04)	−0.41** (0.02)	−0.41** (0.01)
Peer exposure	−4.49** (0.02)	−5.94** (0.03)	−3.18** (0.03)	−6.15** (0.01)	−3.94** (0.01)	−6.83*** (0.01)
School-grade FE	Yes	Yes	Yes	Yes	Yes	Yes
Track indicator	Yes	Yes	Yes	Yes	Yes	Yes
2008 test score controls	Yes	Yes	Yes	Yes	Yes	Yes
Demographic & class controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional classroom controls	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.28	0.29	0.24	0.26	0.36	0.37
Number of observations	1024	1024	1024	1024	1024	1024

Notes: The estimation sample is restricted to students with non-missing outcomes and observed old-school and old-class identifiers. Peer exposure is the leave-one-out classroom share of students whose old school collapsed; own exposure is an indicator for the student's own old-school collapse. All scores are standardized to mean zero and standard deviation one. Column (1) uses the Table 1 column (3) specification, re-estimated on this restricted sample. Both columns include school-grade FE and an advanced-track indicator, the relevant 2008 test score, a missing-score indicator, age, gender, Tibetan minority, and class size. Column (2) additionally includes leave-one-out classroom shares of male and Tibetan-minority classmates, the outcome-specific mean 2008 peer score and its missing-score share, classmates from the same old school, and classmates from the same old school and old class. Wild cluster bootstrap p -values, based on 999 replications, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table B.2 reports a coefficient-stability analysis based on [Oster \(2019\)](#). For each outcome, it compares a parsimonious specification, which includes peer exposure, own

³Peer exposure is only weakly correlated with the shares of classmates from the same old school ($r = -0.23$) and from the same old class ($r = -0.25$), providing limited scope for pre-transition social-network composition to account for the estimated effect.

exposure, school-grade FE, and an advanced-track indicator, with an augmented specification that additionally includes the relevant 2008 test score and missing-score indicator, age, gender, Tibetan minority, class size, and the classroom-composition controls in Appendix Table B.1. Both specifications use the same 1,024 students. The peer-exposure estimate and its Oster lower bound remain negative for all three outcomes.

Table B.2: Coefficient stability of peer-exposure estimates with classroom-composition controls.

	Composition controls	β^*	δ
Chinese	−5.94	−6.50	−10.58
Math	−6.15	−7.80	−3.72
Total	−6.83	−8.32	−4.56

Notes: Each row compares a parsimonious specification with an augmented specification that adds the Table 1 controls and the classroom-composition controls in Appendix Table B.1. Both specifications use the same 1,024 students. *Composition controls* reports the peer-exposure coefficient from the augmented specification. β^* is the Oster lower-bound estimate under proportional selection ($\delta = 1$). δ is the degree of selection on unobservables relative to observables required to drive the peer-exposure effect to zero. The attainable R-squared is set to 1.3 times the R-squared of the augmented specification.

Following [Oster \(2019\)](#), Table B.2 reports two statistics. First, β^* is the implied lower-bound effect when selection on unobservables is assumed to be as strong as selection on observables ($\delta = 1$). Second, δ is the proportional degree of selection on unobservables, relative to observables, that would be required to drive the peer-exposure effect to zero. The attainable R-squared is set to 1.3 times the R-squared of the augmented specification for each outcome.

The augmented composition controls make the peer-exposure estimates more negative for Chinese, Math, and Total score, and the corresponding β^* estimates remain negative. The implied δ values are negative because the observed controls move the coefficient away from zero. Under the proportional-selection calibration, eliminating the estimated effect would therefore require selection on unobservables in the direction opposite to selection on the observed controls. These calculations reinforce the conclusion that the observed classroom composition and measured pre-transition social networks do not account for the negative peer-exposure effects.

C School Resumption, Relocation, and Return

The school questionnaire and teacher questionnaire record the temporary school destination in 2008-09 for each of 22 pre-earthquake Wenchuan schools. They also record the pre- and post-earthquake school names, which permit a school-name link after the return to Wenchuan. Table C.1 summarizes these school-level records.

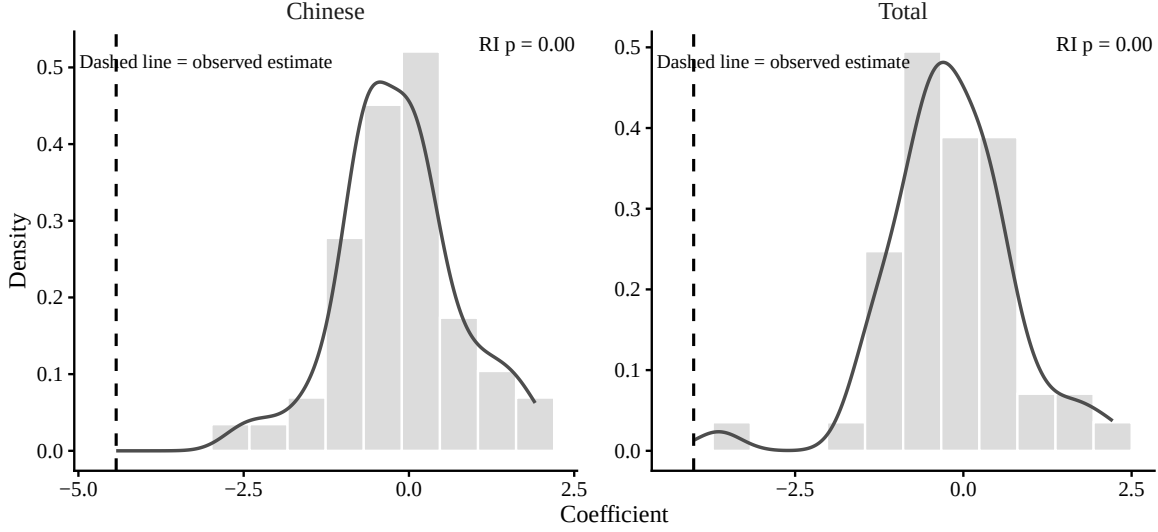


Figure B.1: Placebo classroom-reassignment test for peer exposure.

Notes. Distribution of peer-exposure coefficients from 50 random reallocations of students across observed classrooms within each school-grade cell, using the specification in column (3) of Table 1. The randomization-inference (RI) p -value is the share of placebo estimates at least as large in absolute value as the observed estimate.

During the 2008-09 transition period, 17 of the 22 schools (77.3 percent) resumed instruction outside their original township: 15 in another city in Sichuan and two in another province. The remaining five schools resumed within their original township. This documentation concerns school-level temporary placement, not individual daily commuting, since students moved with their schools during the transition.

For 17 schools, the records identify a post-return Wenchuan school name: 11 retained the same name, while six have a changed name or are recorded as part of a merger. The remaining five have no post-return school name recorded. Importantly, these are school-name matches. They document organizational continuity, renaming, or merger, but do not establish whether the post-return school occupied the same physical site as the pre-earthquake school. Accordingly, the records provide evidence on school resumption and relocation, rather than on individual travel distances after return.

Although official county-level statistics do not separately report permanent schooling-related outmigration from Wenchuan, several institutional features make large-scale educational outmigration unlikely.⁴

⁴China's compulsory education system emphasized nearby exam-free enrollment, typically linked to the student's hukou location, and compulsory education was tuition- and fee-free by this period ([State Council of the People's Republic of China \(2006\)](#); [Sichuan Provincial Department of Education \(2006\)](#)). Wenchuan was also a poor, predominantly ethnic-minority county ([The Central People's Government of the People's Republic of China \(2008\)](#)), making a switch into non-local private schools less plausible. Official reports further indicate that post-earthquake student movement mainly took the form of temporary off-site school resumption.

Table C.1: School-level post-earthquake relocation and return in Wenchuan County.

	Schools	Share
<i>Panel A: Recorded temporary placement, 2008-09</i>		
Other province	2	9.1%
Other city in Sichuan	15	68.2%
Within the original township	5	22.7%
Total	22	100.0%
<i>Panel B: Recorded post-return school-name link</i>		
Same school name	11	50.0%
Changed name or merger	6	27.3%
No post-return school name recorded	5	22.7%
Total	22	100.0%

Notes: Panel A reports the recorded temporary destination for each of the 22 pre-earthquake Wenchuan schools during the 2008-09 transition period. Panel B reports the post-return school-name link in the same records. A school-name link identifies whether the recorded post-return Wenchuan school retained the same name, changed name, or merged with another school; it does not identify whether the school occupied the same physical address before and after the earthquake.

D School-Damage Documentation

To address possible reporting errors in the headmasters' damage classification, I cross-reference the *Local Earthquake Relief Reports on Sichuan Education System*.⁵ Official records identify only two fully collapsed schools, both in Yingxiu Town at the epicenter, where seismic activity was most intense. All other schools in Wenchuan County were reported as severely damaged or partially collapsed, except two primary schools near the epicenter; contemporaneous news reports indicate that their buildings remained relatively unscathed because they had been built more recently and to a higher structural standard.

A second concern is that some students came from old schools outside the surveyed counties, for which damage was not recorded.⁶ For these students, I infer the most probable earthquake intensity by matching each old school's location to the county classification in the *Report on the Assessment of the Scope of the Wenchuan Earthquake Disaster* (Government of China, 2008), which categorizes counties as extremely severely affected, severely affected, or affected. This procedure fills in 247 missing values of old school damage, leaving 70 observations missing because of unclear school names or missing intensity information.

Among the schools surveyed in Wenchuan County, two reported that their buildings did not collapse during the earthquake. Reports related to the Sichuan earthquake confirmed that the damage to these two schools was less severe than that to neighboring schools.

(i) The first school was Caopo Primary School. A report about survivors from Caopo Primary School published in April 2018 in *China News* (Title: One woman's memory of the Sichuan earthquake) attests that the school building did not collapse. The following excerpt is from this report:

Zhou and her younger sister were attending class at Caopo Primary School in the county that day. [Unlike thousands of other buildings in southwest China, Zhou's school somehow avoided collapse]. Zhou and all of her classmates survived the earthquake. But Zhou would soon learn that her parents and elder sister hadn't survived.

(ii) The second school is Baihua Primary School. A report in the *Zhongshan Business Daily*, in September 2008, discussing post-earthquake school reconstruction (Title: New Xuan Kou Primary School to Build the Strongest Structure) mentioned the state of Baihua Primary School. The article also noted that the school building, donated by Mr. Shao Yifu, did not

⁵It is titled *Sichuansheng Jiaoyu Xitong Wenchuan Teda Dizhen Kangzhen Jiuzai Zhi* in Chinese. This report contains records of the extent of damage to school buildings in each county, as maintained by the Sichuan Education Bureau. This report is part of the *Earthquake Relief Reports* collection, which Cao (2022) uses to assess building damage following the Sichuan earthquake.

⁶Out of 1,841 surveyed students, 17% reported an old school name that was not registered among the surveyed schools.

likely collapse because of its high-quality construction. The following is a statement about this construction:

Upon visiting the site, it was observed that the mountainside next to the school had collapsed, and the roads were uneven. A power station near the school had three buildings, one partially collapsed and two completely. The school's gates and walls had fallen, but the main teaching building remained. An adjacent toilet block had entirely collapsed... [The teaching building, constructed in 2002 with a donation of 400,000 yuan from Mr. Shao Yifu of Hong Kong, showed some exterior wall peeling.] On closer inspection, the building bore many scars, and the steel reinforcements in many pillars were twisted and exposed. [Nevertheless, they resiliently supported the building, safeguarding the vibrant lives of hundreds of children.]

E Extended-Sample Evidence

The main analysis focuses on Wenchuan because the county provides the most internally comparable institutional setting and most closely matches the paper’s target population of disaster-affected students. Recovery arrangements differed across counties, and students from old schools that collapsed or sustained severe damage account for 949 of 1,046 students in Wenchuan, compared with 11 of 409 students in Mao and 10 of 568 students in Wangcang. I therefore use Mao and Wangcang as a broader but less homogeneous extension that addresses the limited number of classrooms in the main sample.

Figure E.1 locates the three counties. Wenchuan and Mao contribute 2010 administrative Chinese and math scores, while Wangcang contributes prior-semester scores self-reported in May 2011. Chinese, math, and their raw sum are standardized separately within county and grade so that normalization does not pool unlike score scales.

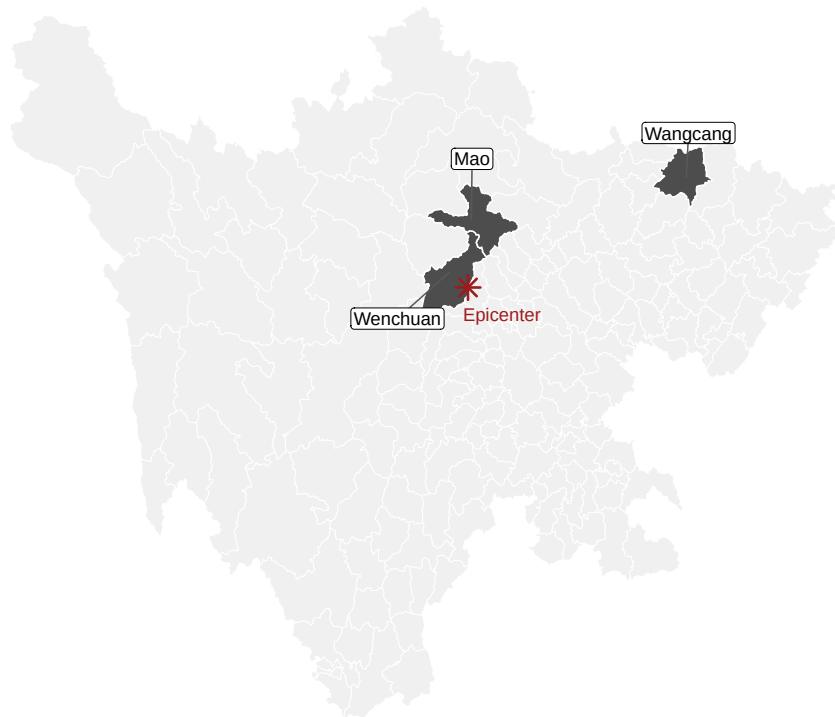


Figure E.1: Location of counties in the extended sample.

Notes. The map shows Sichuan Province. Wenchuan, Mao, and Wangcang are highlighted in dark gray. The star marks the epicenter of the 2008 Wenchuan earthquake.

To verify whether students in the Mao and Wangcang extensions were also randomly assigned across classrooms, I conduct a balance check using the same pairwise procedure applied to the Wenchuan sample in Section A. The full Mao and Wangcang survey roster contains 12 school-grade cells, each with two sampled classrooms. Three Mao cells compare

an advanced class with a regular class and are excluded from unconditional classroom-balance comparisons. Across age, gender, minority status, old-school damage, and home damage, 2 of 41 valid pairwise tests in the remaining cells reject equality at the 5 percent level, or 4.9 percent, and none remains significant after the Benjamini-Hochberg adjustment. Joint 999-draw within-cell permutation tests also fail to reject balance in Mao ($p = 0.41$) or Wangcang ($p = 0.47$). None of the 12 old-school-identity assignment tests rejects at 5 percent.

Figure E.2 displays the within-school-grade classroom variation added by Mao and Wangcang after removing school-grade FE and an advanced-track indicator.

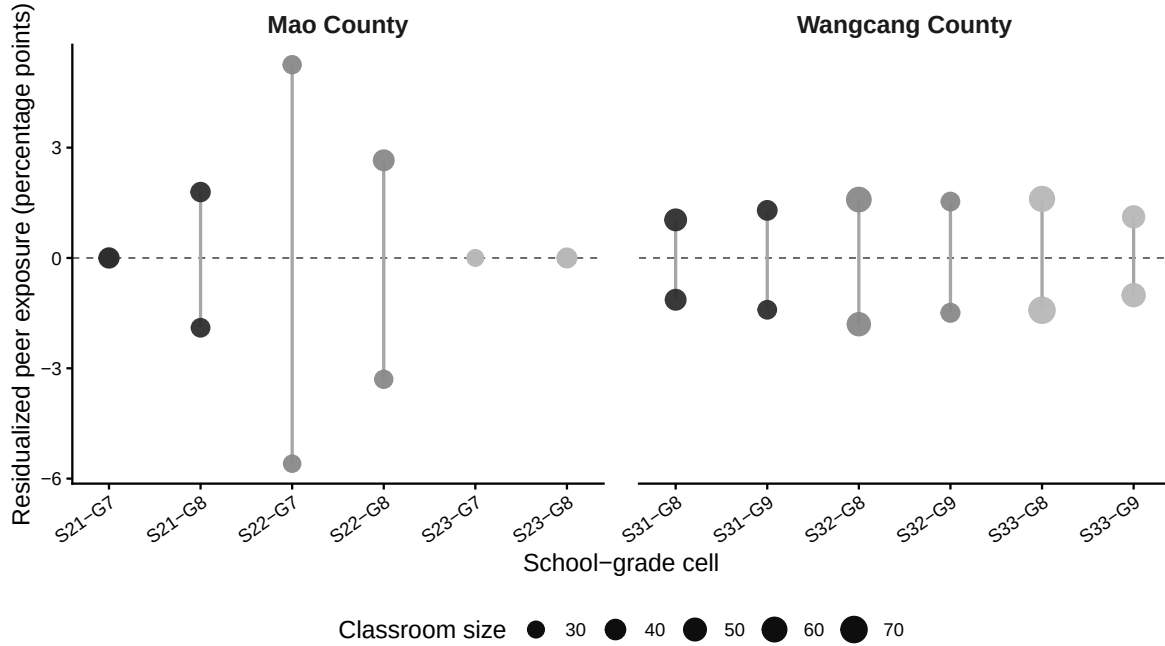


Figure E.2: Residualized variation in classroom peer exposure in Mao and Wangcang.

Notes. Each panel plots classroom means of the leave-one-out share of school-collapsed classmates after residualizing the student-level measure with respect to school-grade FE and an advanced-track indicator. Each column is one school-grade cell, each point is one classroom, vertical segments show the within-cell range, and point size is proportional to classroom size. Mao contributes 409 students in 12 classrooms and six school-grade cells; Wangcang contributes 568 students in 12 classrooms and six school-grade cells. In both counties, old-school damage is the within-old-school median of student reports. School exposure = $\mathbf{1}(\text{Harmonized damage level} \geq 4)$.

E.1 Expanded-Sample Estimates

Table E.1 reports the classroom-level extension.

Table E.1: Effects of own and peer earthquake exposure on test scores: Extended sample.

	Chinese	Math	Total
Own exposure	−0.17** (0.08)	−0.27** (0.12)	−0.26** (0.10)
Peer exposure	−3.02** (1.32)	−4.27*** (1.24)	−4.27*** (1.22)
School-grade FE	Yes	Yes	Yes
Track indicator	Yes	Yes	Yes
Demographic & class size controls	Yes	Yes	Yes
Bootstrap p : own exposure	0.04	0.03	0.02
Bootstrap p : peer exposure	0.10	< 0.01	< 0.01
Adjusted R^2	0.27	0.24	0.27
Number of observations	2023	2023	2023
Number of classroom clusters	51	51	51

Notes: The sample combines 2010 administrative scores from Wenchuan and Mao with prior-semester self-reported scores from Wangcang’s May 2011 survey. Scores are standardized within county and grade. Peer exposure is the leave-one-out classroom share of students whose old school collapsed or sustained severe damage; own exposure is the corresponding student indicator. In Mao and Wangcang, old-school damage is the within-old-school median of student reports. All specifications include school-grade FE and an advanced-track indicator, age, gender, minority status, and class size. Standard errors, clustered at the classroom level, are presented in parentheses. Stars reflect conventional clustered p -values. The two bootstrap rows report impose-null Rademacher wild-cluster p -values based on 999 classroom-level replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The peer-exposure coefficients are −3.02 for Chinese and −4.27 for both math and total score. The corresponding wild-cluster bootstrap p -values are 0.10, less than 0.01, and less than 0.01. Thus, the negative classroom peer-exposure pattern remains present in the larger sample, although the extension combines less comparable institutional and outcome-measurement settings.

Table E.2: Effects of own and grade-level peer earthquake exposure on test scores: Extended sample.

	Chinese	Math	Total
Own exposure	−0.17* (0.08)	−0.26** (0.07)	−0.25** (0.08)
Grade-level peer exposure	−1.22 (0.43)	−1.34 (0.60)	−1.29 (0.48)
School FE	Yes	Yes	Yes
Grade FE	Yes	Yes	Yes
Demographic & class size controls	Yes	Yes	Yes
Bootstrap p : own exposure	0.09	0.01	0.03
Bootstrap p : grade-level peer exposure	0.16	0.42	0.33
Adjusted R^2	0.23	0.15	0.18
Number of observations	2023	2023	2023
Number of school-grade clusters	22	22	22

Notes: The sample, outcomes, exposure definitions, and demographic and class-size controls follow Table E.1. Peer exposure is computed over students in the same school-grade cell rather than the classroom. Because this measure varies at the school-grade level, the specification uses additive school and grade FE, without the advanced-track indicator or a 2008 test score. Standard errors, clustered at the school-grade level, are presented in parentheses. Stars and the two bootstrap rows reflect impose-null Rademacher wild-cluster p -values based on 999 school-grade-level replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.2 replaces classroom peer exposure with the leave-one-out exposure share in the school-grade cell. Because this measure varies across school-grade cells rather than across classrooms within a cell, the specification uses additive school and grade FE. The coefficients remain negative, but the school-grade-level wild-cluster bootstrap p -values are 0.16 for Chinese, 0.42 for math, and 0.33 for total score. The broader grade-level estimates are therefore smaller and less precise than the classroom-level estimates.

F Peer Effects by Own School-Collapse Status

The full-sample estimates in Figure 3 pool two effects of school-collapsed classmates: on students whose own schools collapsed, the intragroup peer effect, and on students whose own schools did not, the intergroup peer effect. Because the former is the paper’s central interest, I estimate peer effects separately by own school-collapse status. Appendix Figure K.6 restricts the focal sample to students whose own schools collapsed, and Appendix Table J.8 interacts peer exposure with own school-collapse status. The main purpose is to recover the magnitude of the intragroup peer effect.

Among students whose own schools collapsed, peer exposure significantly lowers all three

academic outcomes, happiness, and self-rated health. These estimates are similar to the full-sample estimates and somewhat larger in absolute value than those for non-collapsed students. The group differences, however, are not statistically distinguishable from zero, and the comparison is imprecise because non-collapsed students make up only 9.3 percent of the main estimation sample. The evidence therefore does not establish differential effects by own school-collapse status, although the larger intragroup magnitudes are consistent with a negative spiral in which shared severe disaster experience amplifies adverse peer effects.

G Construction of Baseline Rank

Baseline rank is constructed from the 2008 post-earthquake test scores, recorded while students were still enrolled in their old schools. For survey grades 8 and 11, this was the final-year examination in the old school; for grades 7 and 10, it preceded the final year by one academic year. In all cases, the scores predate enrollment in the new school and classroom assignment, so baseline rank is predetermined with respect to peer composition. The remaining concern is contamination: if severe old-school damage had already depressed 2008 scores, heterogeneity by “baseline rank” could partly sort students by earthquake severity rather than prior academic standing. I address this concern in two ways. First, regressions of 2008 achievement on old-school damage, with the standard controls, school-grade FE, and an advanced-track indicator, show no evidence that severe damage reduced baseline performance: the own-exposure coefficient is 0.10 standard deviations for the 2008 total score ($p = 0.38$) and is likewise small and insignificant for 2008 language, math, and baseline rank. Second, I define rank using buffered extreme groups rather than a hard split. Within each school-grade, the bottom 40 percent are classified as low-rank and the top 40 percent as high-rank, leaving the middle 20 percent unlabeled, which keeps the comparison away from students close to the cutoff.⁷ Because the 40 percent cutoff is arbitrary, Appendix Figure K.7 varies it from 35 to 45 percent.

H Comparison with Related Evidence on Gender

A related study in the Sichuan earthquake setting finds that earthquake exposure benefits girls’ long-run educational attainment but has no significant effect on boys (Liu and Xu, 2021). This finding does not contradict mine, because the treatments and time horizons differ. Liu and Xu estimate a county-level effect that combines the earthquake with subsequent

⁷As a simple back-of-the-envelope calculation intended to show that false classification is very limited under the buffered group definition, I assume that the effect of own exposure on test scores is 0.41 standard deviations, as estimated in Table 1, add 0.41 back to the 2008 total score of students whose own old school collapsed, and then re-rank students. Under this adjustment, no students under the school-grade 40/40 split move from the calibrated high-rank group into the observed low-rank group.

school reconstruction and fee relief. I instead study academic performance about two years after the earthquake in Wenchuan County, the epicenter, where approximately 94% of school buildings were damaged. Within this uniformly hard-hit setting, my direct-effect estimates compare partial or full old-school collapse (damage levels 4 or 5) with less extensive damage (levels 1 to 3), capturing the incremental short-run effect of more severe school-level exposure. Moreover, the gender difference in my setting appears not in the direct effect, where male and female losses are statistically indistinguishable, but in the classroom peer effect among students whose own schools collapsed. Section 7.1 examines whether gendered friendship and relational processes help explain this larger adverse peer effect for female students.

I Alternative Explanations

I.1 Teacher Pedagogical Practices

Teachers may adjust their time allocation in response to classroom earthquake exposure. Because the matched teacher survey covers only 18 head teachers,⁸ I treat this evidence as descriptive. Head teachers in more exposed classrooms appear to spend more time on student life support, with little change in teaching preparation. Preparation time is positively but insignificantly associated with Chinese and math scores, whereas life-support time shows no association with achievement (Appendix Table J.11). The evidence therefore does not support endogenous pedagogical adjustment as an explanation.

I.2 Time Allocation and Study Effort

If classrooms with more affected classmates are less conducive to learning, students may disengage from schoolwork outside school. Appendix Figure K.10 shows no such decline: the full-sample estimate for after-school study time is positive and insignificant, and no subgroup shows reduced effort. If anything, high-rank students study more as peer exposure rises (coefficient = 3.51, $p = 0.10$). This marginally significant pattern is consistent with a compensatory response and may help explain why high-rank students avoid comparable academic losses.

I.3 Teacher Quality, Teacher Assignment, and Classroom Inputs

Peer exposure could also proxy for teacher-side differences, which I examine in three ways. First, schools that avoided collapse may have had stronger teachers before the earthquake. However, the share of high-title teachers in the student's pre-earthquake school does not

⁸Teachers' self-reported hours for teaching preparation and student support are converted into percentage ranks to mitigate reporting errors and outliers.

predict old-school collapse ($\hat{\beta} = -0.08$, $p = 0.61$; Appendix Table J.12). Second, stronger head teachers may have been assigned to less-exposed classrooms after resettlement. In the 26 classrooms with matched teacher data, classroom exposure does not predict a high-title head teacher in either the Chinese ($\hat{\beta} = -0.33$, $p = 0.77$) or the math sample ($\hat{\beta} = 0.28$, $p = 0.83$), although the small sample limits this test. Third, adding the old-school teacher-title share, the current head teacher’s title, and teaching time, separately or jointly, barely moves the estimates (Appendix Figure K.11): the peer-exposure coefficient ranges from -4.28 to -5.00 for Chinese, -3.11 to -3.60 for math, -3.79 to -4.46 for total score, and 1.80 to 2.19 for SDQ difficulties. Teacher quality, teacher assignment, and these observed inputs therefore do not drive the main results.

J Additional Tables

Table J.1: Summary statistics.

	Completion	Mean	Std. Dev.	Median	Min.–Max.
<i>Panel A: Full Sample ($N = 1278$)</i>					
Chinese score (2010)	0.88	64.14	17.09	65.00	8.00–100.00
Math score (2010)	0.85	39.64	25.45	33.00	2.00–100.00
Standardized Chinese score (2010)	0.88	0.00	1.00	0.10	-3.46–2.18
Standardized math score (2010)	0.85	0.00	1.00	-0.19	-1.84–3.30
Male	1.00	0.50	0.50	1.00	0.00–1.00
Age	0.99	13.72	1.31	14.00	10.00–18.00
Ethnic minority	1.00	0.77	0.42	1.00	0.00–1.00
Tibetan minority	1.00	0.27	0.45	0.00	0.00–1.00
Parental education	0.99	0.33	0.47	0.00	0.00–1.00
Earthquake exposure	0.97	0.91	0.29	1.00	0.00–1.00
Old school damage level	0.97	3.89	0.62	4.00	2.00–5.00
<i>Panel B: Estimation Sample (Students with Non-Missing Administrative Scores) ($N = 1086$)</i>					
Chinese score (2010)	1.00	63.60	16.87	65.00	8.00–99.00
Math score (2010)	1.00	39.56	25.40	33.00	2.00–100.00
Standardized Chinese score (2010)	1.00	-0.03	0.99	0.10	-3.46–2.18
Standardized math score (2010)	1.00	-0.00	1.00	-0.19	-1.84–3.30
Male	1.00	0.48	0.50	0.00	0.00–1.00
Age	0.99	13.69	1.30	14.00	10.00–18.00
Ethnic minority	1.00	0.77	0.42	1.00	0.00–1.00
Tibetan minority	1.00	0.28	0.45	0.00	0.00–1.00
Parental education	0.99	0.31	0.46	0.00	0.00–1.00
Earthquake exposure	0.97	0.91	0.29	1.00	0.00–1.00
Old school damage level	0.97	3.90	0.63	4.00	2.00–5.00

Notes: Panel B restricts the full sample in Panel A to students with non-missing 2010 administrative Chinese and math scores. Both panels exclude grades 10 and 11 at one six-year secondary school without administrative records. Standardized scores have mean zero and unit standard deviation. Ethnic minority = 1 if non-Han; Tibetan minority = 1 if Tibetan, the ethnicity control in all regressions. Parental education = 1 if at least one parent completed high school or above. Old school damage ranges from 1 (intact) through 2 (moderate damage), 3 (severe damage), and 4 (partial collapse) to 5 (full collapse); no old school is intact. Earthquake exposure = 1 (old school damage ≥ 4). Psychosocial summary statistics are in Appendix Table J.2.

Table J.2: Summary statistics for psychosocial outcomes.

	Completion	Mean	Std. Dev.	Median
<i>Panel A: Full Sample ($N = 1278$)</i>				
CES-D (raw sum)	0.95	34.41	5.12	34.00
CES-D (standardized)	0.95	0.00	1.00	-0.08
SDQ (raw sum)	0.97	23.94	2.16	24.00
SDQ (standardized)	0.97	-0.00	1.00	0.03
Rosenberg (raw sum)	0.96	25.33	2.47	26.00
Rosenberg (standardized)	0.96	-0.00	1.00	-0.27
FES (raw sum)	0.96	26.57	1.75	27.00
FES (standardized)	0.96	0.00	1.00	-0.24
<i>Panel B: Estimation Sample (Students with Non-Missing Administrative Scores) ($N = 1086$)</i>				
CES-D (raw sum)	0.95	34.48	5.14	34.00
CES-D (standardized)	0.95	-0.00	1.00	-0.09
SDQ (raw sum)	0.98	23.95	2.15	24.00
SDQ (standardized)	0.98	-0.00	1.00	0.03
Rosenberg (raw sum)	0.97	25.32	2.49	26.00
Rosenberg (standardized)	0.97	0.00	1.00	-0.27
FES (raw sum)	0.96	26.53	1.72	27.00
FES (standardized)	0.96	0.00	1.00	-0.27

Notes: The table reports both raw-sum and standardized summary statistics for the four psychosocial scales in the survey. CES-D is the Center for Epidemiologic Studies Depression Scale; 12 of the original 20 items are available, and higher values indicate more depressive symptoms. SDQ is the Strengths and Difficulties Questionnaire; only the conduct and peer-problems subscales are observed (10 items total), and higher values indicate greater difficulties. Rosenberg is the Rosenberg Self-Esteem Scale; all 10 items are included, and the standardized measure is reverse-coded so that higher values indicate lower self-esteem. FES is the Family Environment Scale; 18 items are available, and higher values indicate more family conflict. Standardized rows are normalized to mean zero and unit standard deviation within the corresponding sample.

Table J.3: Pairwise balance tests within school-grade cells, with pairs formed separately by track.

Variable	Valid t-tests	Sig. at 5%
Own old-school collapse	17	0 (0.0%)
Own old-school damage scale	17	0 (0.0%)
Age	21	0 (0.0%)
Gender	21	4 (19.0%)
Tibetan minority	20	1 (5.0%)
Parental education	21	2 (9.5%)
All five characteristics	100	7 (7.0%)

Notes: All feasible classroom pairs are formed within each school-grade cell, separately by track in the one tracked school. The table reports the number of valid pairwise t-test comparisons and the number significant at the 5% level. “Own old-school damage scale” is the original five-point damage measure from the student survey; “Own old-school collapse” is the derived binary indicator. Tibetan minority equals 1 for students of Tibetan ethnicity. “All five characteristics” pools age, gender, Tibetan minority, parental education, and the old-school damage scale.

Table J.4: Predetermined characteristics and earthquake exposure.

	Old school damage	Earthquake exposure
	(1)	(2)
Distance from epicenter (10 km)	−0.002 (0.978)	0.030 (0.471)
Age	0.145 (0.112)	0.040 (0.304)
Male	−0.085 (0.190)	−0.025 (0.538)
Tibetan minority	−0.397 (0.489)	−0.157 (0.744)
Rural hukou	−0.079 (0.436)	−0.043 (0.288)
Parental education	0.096 (0.300)	0.036 (0.541)
Pre-earthquake wealth	0.019 (0.368)	0.006 (0.726)
Household size	−0.006 (0.624)	0.002 (0.747)
Lives in county seat	0.028 (0.698)	−0.004 (0.814)
Joint wild-bootstrap p -value	0.963	0.999
Characteristics in joint test	8	8
Adjusted R^2	0.148	0.131
Number of observations	876	876
Old schools	22	22

Notes: Column (1) uses old school damage, a five-point scale: 1 (intact), 2 (moderate damage), 3 (severe damage), 4 (partial collapse), and 5 (full collapse). Column (2) uses earthquake exposure, an indicator equal to one when old school damage is 4 or 5. Higher values in both columns indicate greater earthquake exposure. The sample includes Wenchuan students with non-missing values for the two dependent variables, old-school distance, and all eight predetermined characteristics listed in the table. The regressions include no FE. The joint tests exclude distance and test only the displayed predetermined-characteristic block. Distance from the epicenter has limited predictive power for school damage within Wenchuan. One possible reason is that Wenchuan is a single county-level administrative unit covering 4,084 km² that contains the earthquake epicenter and was among the counties most severely affected by the earthquake. When more distant old schools in Mao and Wangcang are added in separate descriptive checks, the distance coefficients are −0.068 for old school damage and −0.031 for earthquake exposure per 10 km ($p < 0.001$ in both cases). These pooled gradients are cross-county and should not be interpreted as within-county evidence. Validated distance is available for 22 of 33 harmonized old-school codes. The remaining 11 old-school codes lack an old-school GIS coordinate match. Wild cluster bootstrap p -values, based on 999 old-school-level replications, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.5: Placebo regressions for predetermined student characteristics.

Outcome	(1) Coef.	(2) p -value	(3) Number of observations
Age	0.76	0.20	1046
Male	−0.88	0.08	1046
Tibetan minority	−0.35	0.44	1046
Rural hukou	0.63	0.14	1033
Parental education	−0.56	0.11	1041
Pre-earthquake wealth	−2.49	0.17	1046
Household size	0.07	0.79	1046
Lives in county seat	−0.21	0.74	1043

Notes: Each row treats a predetermined student characteristic as the outcome in the whole-sample specification with peer exposure and own exposure as regressors. The table reports only the coefficient on peer exposure and its p -value. All regressions include school-grade FE and an advanced-track indicator, and are clustered at the classroom level. Reported p -values are from impose-null Rademacher wild cluster bootstrap inference with 999 classroom-level replications. Sample sizes vary because some background characteristics are missing for a subset of students.

Table J.6: Alternative clustering of own- and peer-exposure estimates.

Outcome	Exposure	Coefficient	Wild-bootstrap p -value	
		(1)	(2) Classroom clusters	(3) School-grade clusters
Chinese	Peer exposure	−4.43	0.02	0.07
	Own exposure	−0.33	0.00	0.03
Math	Peer exposure	−3.31	0.02	0.07
	Own exposure	−0.38	0.02	0.02
Total	Peer exposure	−4.01	0.01	0.06
	Own exposure	−0.41	0.01	0.04

Notes: Column (1) reports the own- and peer-exposure coefficients from column (3) of Table 1. Columns (2) and (3) report impose-null Rademacher wild cluster bootstrap p -values based on 999 replications at the classroom level (27 clusters) and school-grade level (10 clusters), respectively. All specifications retain the Table 1 column (3) sample, controls, school-grade FE, and an advanced-track indicator.

Table J.7: Multiple-testing adjustments for peer effects in Figure 3.

Outcome	Peer-exposure estimate (SE)	p	BH q
<i>Panel A: Psychosocial composites</i>			
Depression (CES-D)	0.43 (0.75)	0.62	0.73
SDQ difficulties	1.79* (0.72)	0.07	0.15
Rosenberg self-esteem	-0.71 (0.71)	0.43	0.61
Family environment (FES)	-0.70 (1.40)	0.80	0.80
<i>Panel B: Wellbeing and motivation</i>			
Happiness	-2.44*** (0.44)	<0.01	<0.01
Self-rated health	-4.37*** (1.14)	<0.01	0.02
Study motivation	-2.88 (1.28)	0.14	0.25

Notes: Each row reports the current Figure 3 peer-exposure coefficient from a separate full-sample regression. Own exposure is included jointly. Estimates are followed by classroom-clustered standard errors in parentheses. The p column reports impose-null Rademacher wild-cluster bootstrap p -values based on 999 classroom-level replications. BH q reports the Benjamini-Hochberg false-discovery-rate adjustment. Adjustments are calculated across the seven peer-exposure outcomes shown in Figure 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ refer to the unadjusted wild-bootstrap p -values. All regressions include school-grade FE, an advanced-track indicator, age, gender, Tibetan minority, and class size.

Table J.8: Peer-exposure effects by students' own school-collapse status.

Outcome	Non-collapsed students	School-collapsed students	Difference	<i>N</i>
<i>Panel A: Academic outcomes</i>				
Chinese	−3.84** (0.02)	−4.59** (0.01)	−0.75 (0.10)	1046
Math	−2.50 (0.21)	−3.53*** (0.01)	−1.03 (0.20)	1046
Total	−3.24** (0.04)	−4.22*** (0.01)	−0.99 (0.15)	1046
<i>Panel B: Psychosocial outcomes</i>				
Depression	0.73 (0.59)	0.36 (0.76)	−0.38 (0.85)	995
SDQ difficulties	0.94 (0.35)	2.02* (0.05)	1.08 (0.16)	1025
Rosenberg self-esteem	−1.67 (0.21)	−0.44 (0.65)	1.22 (0.36)	1010
FES	0.49 (0.83)	−0.98 (0.69)	−1.47** (0.02)	1009
<i>Panel C: Subjective wellbeing and motivation</i>				
Happiness	−1.65** (0.04)	−2.64*** (0.00)	−0.99 (0.20)	1042
Health	−3.43** (0.03)	−4.62*** (0.01)	−1.18 (0.35)	1045
Study motivation	−3.32* (0.08)	−2.77 (0.15)	0.55 (0.46)	1036
School-grade FE		Yes		
Track indicator		Yes		
Demographic & class size controls		Yes		

Notes. Each row represents one separate regression, with the outcome indicated in the first column. Impose-null Rademacher wild-cluster bootstrap p -values are in parentheses. Non-collapsed students are students whose own old school did not collapse; school-collapsed students are students whose own old school collapsed. Difference is the school-collapsed-student peer effect minus the non-collapsed-student peer effect. All specifications include school-grade FE and an advanced-track indicator, age, gender, Tibetan minority, and class size. Academic-outcome specifications additionally control for the relevant 2008 test score and a missing-score indicator. Bootstrap inference uses 999 classroom-level replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.9: Multiple-testing adjustments for heterogeneous peer-exposure estimates.

Outcome	Group or contrast	Peer-exposure estimate (SE)	p	BH q
<i>Panel A: Prior-rank group estimates</i>				
Chinese score	Low-rank	−4.93** (1.38)	0.01	0.06
Chinese score	High-rank	−2.11 (1.24)	0.23	0.31
Math score	Low-rank	−2.57 (1.37)	0.10	0.16
Math score	High-rank	−0.27 (1.42)	0.88	0.91
Total score	Low-rank	−3.44** (1.24)	0.02	0.07
Total score	High-rank	−1.05 (1.14)	0.42	0.48
Overall rank (pctile)	Low-rank	−1.24** (0.36)	0.03	0.07
Overall rank (pctile)	High-rank	0.04 (0.28)	0.91	0.91
<i>Panel B: Prior-rank interaction tests</i>				
Chinese score	High minus low	2.83** (0.80)	0.02	0.03
Math score	High minus low	2.30** (0.56)	0.02	0.03
Total score	High minus low	2.40** (0.58)	0.01	0.03
Overall rank (pctile)	High minus low	1.28*** (0.23)	<0.01	0.02

Notes: Panels A and B report the group-specific peer-exposure coefficients and the high-minus-low interaction coefficients from Panel B of Table 2. Estimates are followed by classroom-clustered standard errors in parentheses. The p column reports impose-null Rademacher wild-cluster bootstrap p -values based on 999 classroom-level replications. BH q reports the Benjamini-Hochberg false-discovery-rate adjustment calculated jointly across the corresponding peer-exposure estimates or interaction tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ refer to the unadjusted wild-bootstrap p -values. All regressions use the corresponding specifications in Table 2.

Table J.9: Multiple-testing adjustments for heterogeneous peer-exposure estimates (continued).

Outcome	Group or contrast	Peer-exposure estimate (SE)	p	BH q
<i>Panel C: Gender group estimates</i>				
Chinese score	Male	−3.48** (1.18)	0.04	0.09
Chinese score	Female	−5.09** (1.29)	0.01	0.06
Math score	Male	−1.67 (1.48)	0.35	0.43
Math score	Female	−2.58 (1.61)	0.14	0.21
Total score	Male	−2.23* (1.22)	0.09	0.15
Total score	Female	−3.62** (1.31)	0.02	0.06
Overall rank (pctile)	Male	−0.69** (0.34)	0.04	0.09
Overall rank (pctile)	Female	−1.13** (0.38)	0.02	0.06
<i>Panel D: Gender interaction tests</i>				
Chinese score	Female minus male	−1.61*** (0.42)	<0.01	0.02
Math score	Female minus male	−0.90 (0.58)	0.28	0.28
Total score	Female minus male	−1.39** (0.32)	0.03	0.03
Overall rank (pctile)	Female minus male	−0.44* (0.12)	0.06	0.07

Notes: Panels C and D report the group-specific peer-exposure coefficients and the female-minus-male interaction coefficients from Panel D of Table 2. Estimates are followed by classroom-clustered standard errors in parentheses. The p column reports impose-null Rademacher wild-cluster bootstrap p -values based on 999 classroom-level replications. BH q reports the Benjamini-Hochberg false-discovery-rate adjustment calculated jointly across the corresponding peer-exposure estimates or interaction tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ refer to the unadjusted wild-bootstrap p -values. All regressions use the corresponding specifications in Table 2.

Table J.10: Multiple-testing adjustments for SDQ subscale peer-exposure estimates.

Outcome	Peer-exposure estimate (SE)	p	BH q
SDQ conduct problems	0.70 (1.03)	0.59	0.59
SDQ peer problems	3.64** (1.46)	0.04	0.08

Notes: The table reports the peer-exposure coefficients for the SDQ subscales in Figure 4. Estimates are followed by classroom-clustered standard errors in parentheses. The p column reports impose-null Rademacher wild-cluster bootstrap p -values based on 999 classroom-level replications. BH q reports the Benjamini-Hochberg false-discovery-rate adjustment across the two SDQ subscale estimates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ refer to the unadjusted wild-bootstrap p -values. All regressions include school-grade FE, an advanced-track indicator, age, gender, Tibetan minority, and class size.

Table J.11: Effects of teaching preparation time and life-support time on academic performance.

	Chinese		Math	
	(1)	(2)	(3)	(4)
Teaching preparation	0.99 (0.24)		1.23 (0.40)	
Life support		−0.31 (0.99)		0.03 (1.00)
School-grade FE	Yes	Yes	Yes	Yes
Track indicator	Yes	Yes	Yes	Yes
Full controls	Yes	Yes	Yes	Yes
R2 Adj.	0.33	0.30	0.26	0.22
Number of observations	773	733	773	733

Notes: The dependent variables are standardized administrative test scores in 2010. Full controls include peer old-school exposure, own old-school exposure, the relevant 2008 test score, a missing-score indicator, age, gender, Tibetan minority, and class size. School-grade FE and an advanced-track indicator are included in all specifications. Wild cluster bootstrap p -values, based on 999 classroom-level Rademacher replications, are in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$ are based on those bootstrap p -values.

Table J.12: Old-school teacher quality and school collapse.

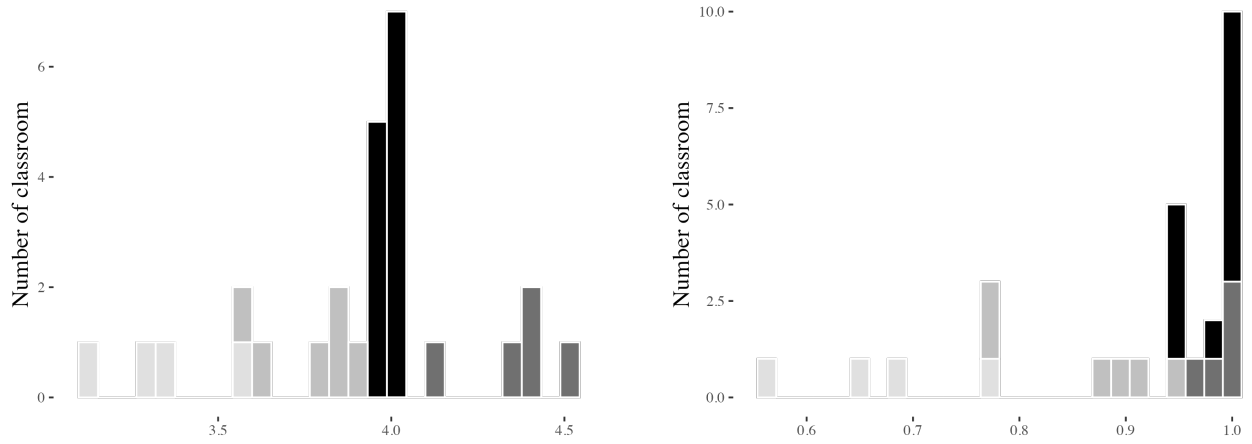
	Old school damage
Intercept	0.94*** (0.00)
Share of high-title teachers	−0.08 (0.61)
Missing indicator	−0.08 (0.55)
R2 Adj.	0.00
Number of Observations	1046

Notes: The dependent variable is an indicator for old-school collapse. The key regressor is the share of high-title teachers in the old school, with a missing-data indicator included separately. Wild cluster bootstrap p -values, based on 999 old-school-level Rademacher replications, are in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$ are based on those bootstrap p -values.

K Additional Figures

Figure K.1 shows the distribution across classrooms of the classroom average of the five-point damage scale (Figure K.1a) and of the binary exposure measure (Figure K.1b), with colors denoting schools. Two of the four schools enrolled students mainly from Wenchuan County, whereas the other two drew a more diverse population from adjacent counties. Appendix Figure K.3 shows the within-school-grade identifying variation across classrooms by plotting peer exposure residualized on school-grade FE and an advanced-track indicator. The 24 classrooms in seven school-grade cells provide variation in peer exposure, whereas the three single-classroom school-grade cells provide none.

Figure K.1: Distribution of treatment measures.

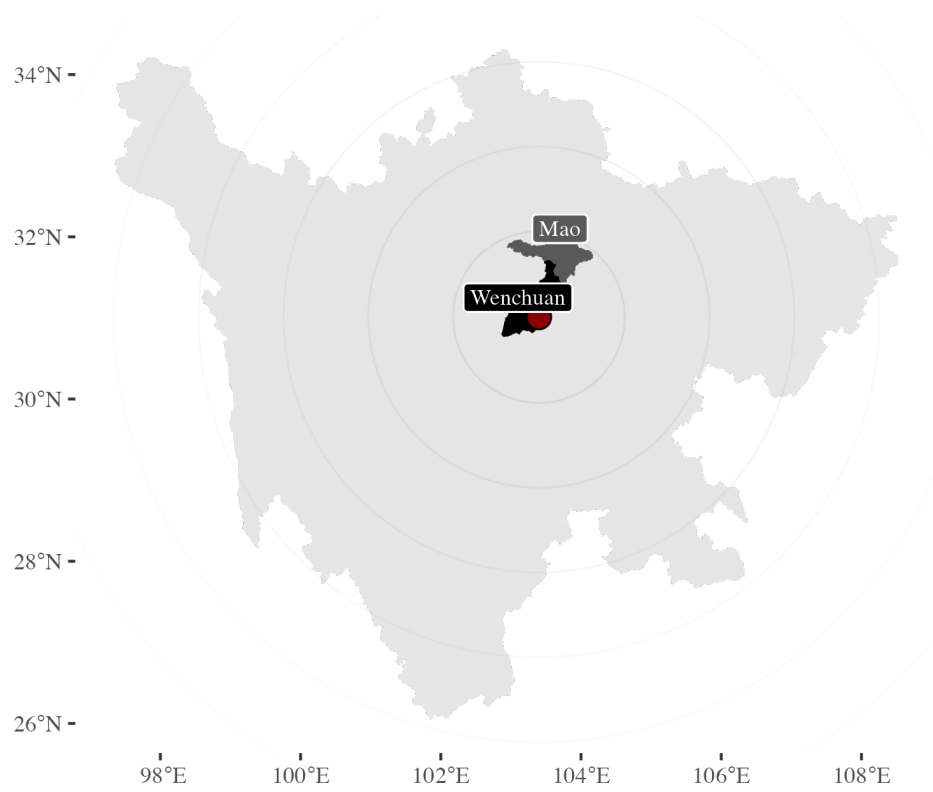


(a) Classroom average old school damage.

(b) Classroom average earthquake exposure.

Notes: Earthquake exposure = $\mathbb{1}(\text{Old school damage level} \geq 4)$. Different colors represent different schools.

Figure K.2: Location of surveyed counties in Sichuan Province.



Notes: The map shows Sichuan Province with the two surveyed counties, Wenchuan and Mao, highlighted.

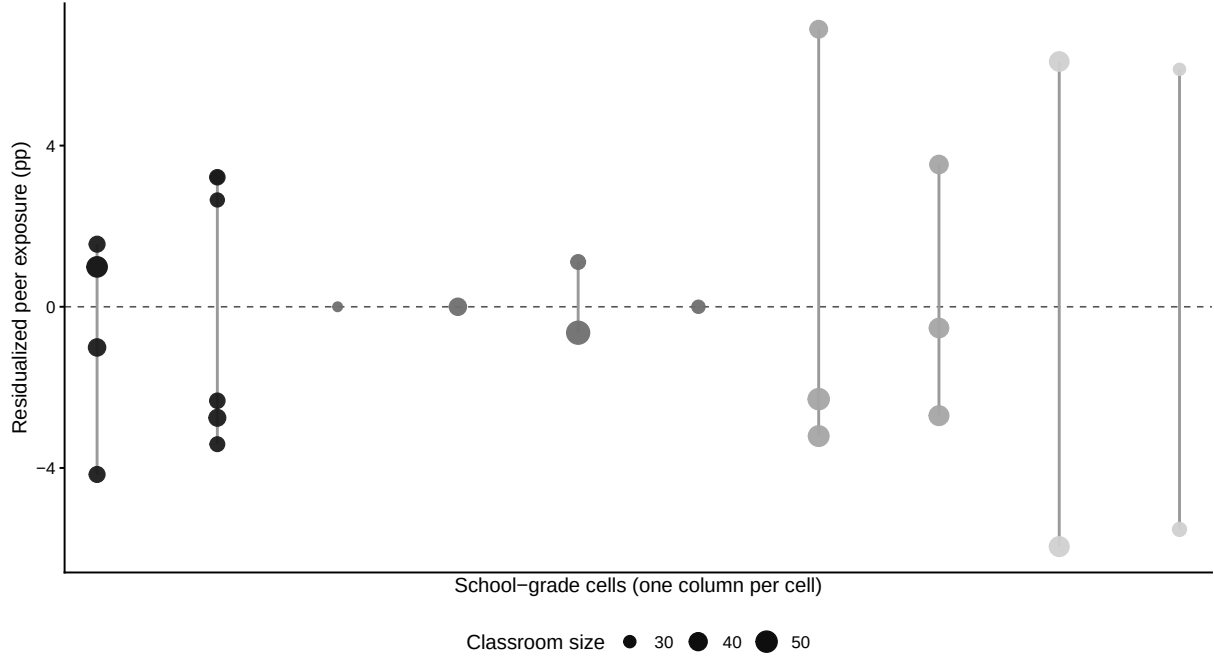


Figure K.3: Residualized variation in classroom peer exposure.

Notes: The figure plots classroom means of the leave-one-out share of school-collapsed classmates after residualizing the student-level measure with respect to school-grade FE and an advanced-track indicator. Each vertical column represents one school-by-grade cell, and different shades distinguish schools. Each point represents one classroom; vertical segments show the within-cell range, and point size is proportional to classroom size. The vertical axis reports residualized peer exposure in percentage points. The estimation sample contains 1,046 students in 27 classrooms and 10 school-by-grade cells. Cells with one classroom provide no within-cell classroom-composition comparison. Earthquake exposure = $\mathbf{1}(\text{Old school damage level} \geq 4)$.

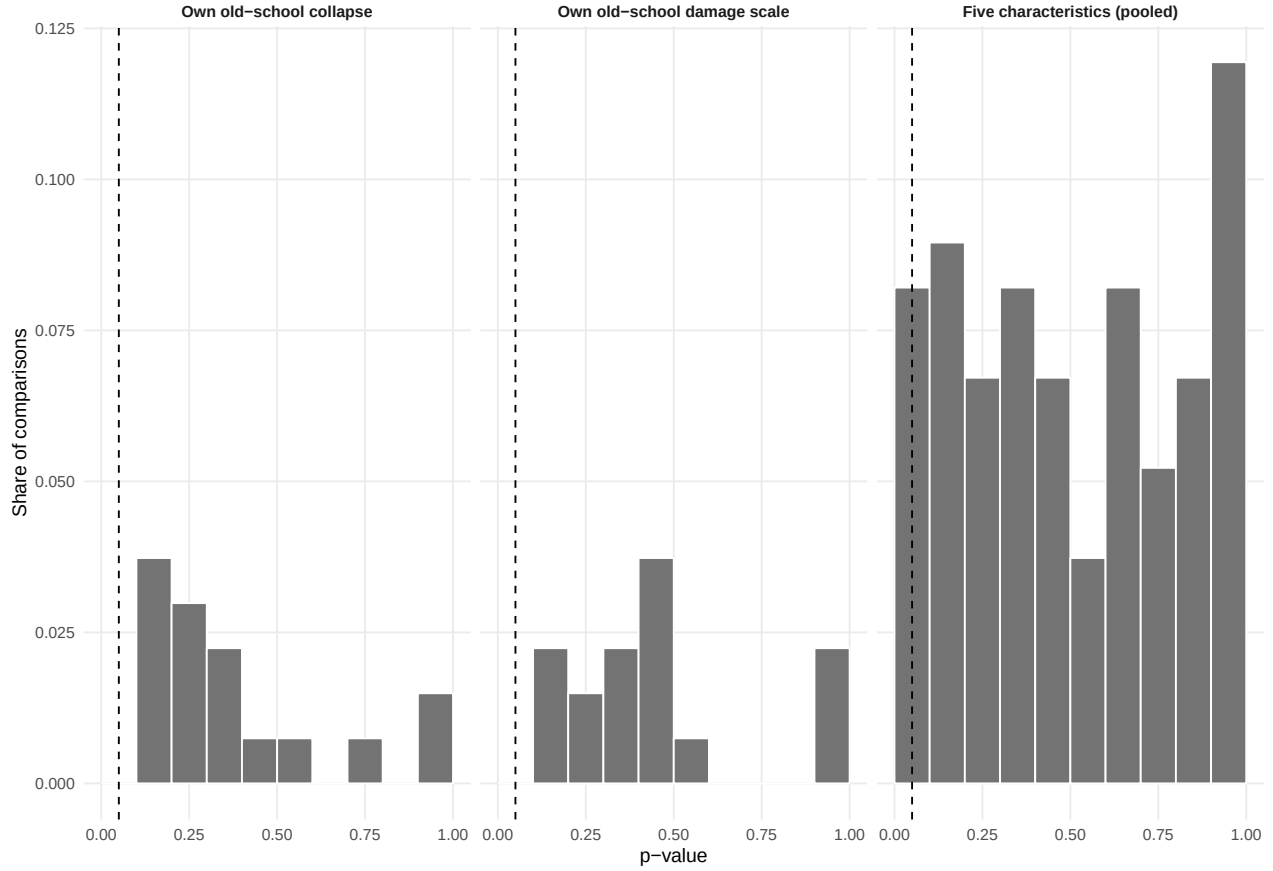


Figure K.4: Distributions of pairwise-balance p-values.

Notes. The figure reports p-value histograms for pairwise t-tests within school-grade cells, with tests formed separately by track in the one tracked school. Panels distinguish the student's own old-school collapse indicator, the corresponding five-point old-school damage scale, and the pooled set of five characteristics: age, gender, Tibetan minority, parental education, and the old-school damage scale. The collapse indicator is derived from the damage scale. The dashed line marks 0.05.

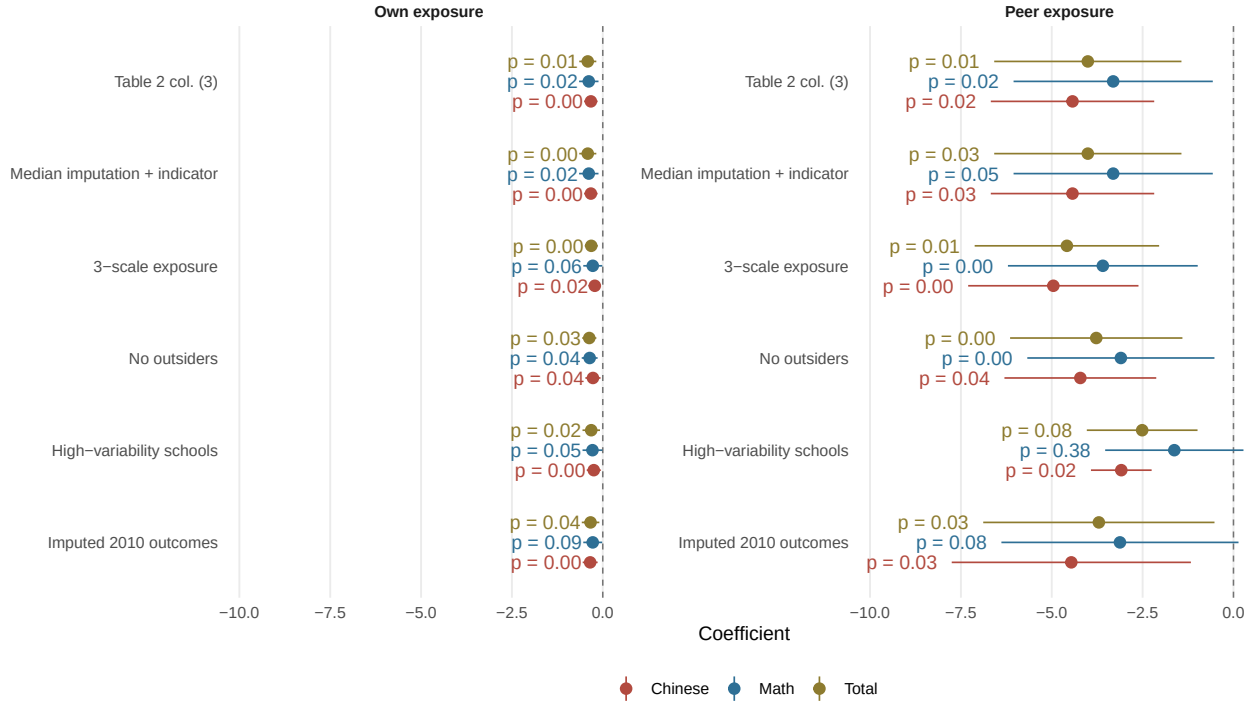
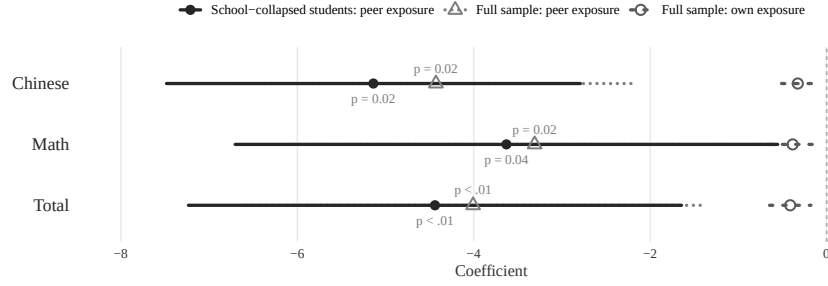
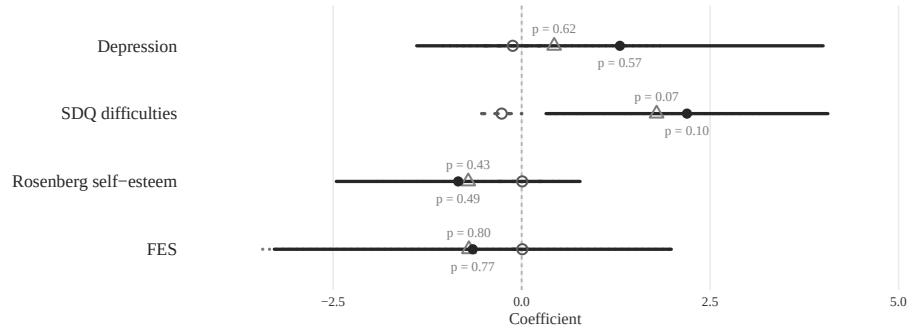


Figure K.5: Sensitivity of peer-exposure estimates to alternative definitions and sample coverage.

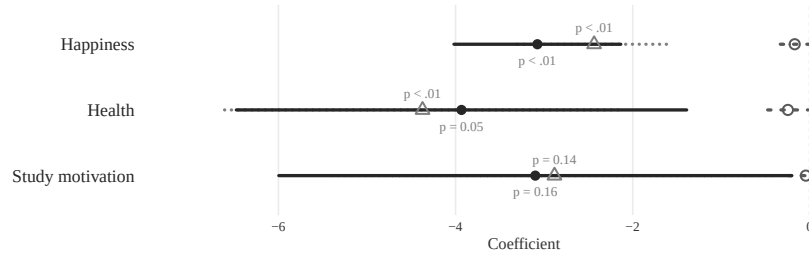
Notes. Each panel reports the coefficients on own exposure and peer exposure from variants of the Table 1 column (3) specification. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. Labels to the left of the confidence intervals report p-values from wild cluster bootstrap inference. “Median imputation + indicator” replaces missing 2008 test scores with the subject-specific sample median and retains the corresponding missing-score indicator. “3-scale exposure” replaces the collapse dummy with a three-level exposure measure based on old-school damage severity. “No outsiders” restricts the sample to students whose old school is observed in the survey data. “High-variability schools” restricts the sample to schools with greater within-school variation in exposure. “Imputed 2010 outcomes” begins with the same 1,278-student full questionnaire sample, replaces missing raw 2010 administrative Chinese and Math scores with the corresponding self-reported raw scores when available, standardizes the combined subject outcomes within grade, reconstructs Total score using the Table 1 rule, recomputes peer exposure on the combined-outcome sample, and re-estimates the fully controlled specification on 1,200 students in the original 27 classrooms. The Pearson correlation between self-reported and administrative raw scores is 0.45 for Chinese and 0.43 for Math.



(a) Academic outcomes



(b) Psychosocial outcomes

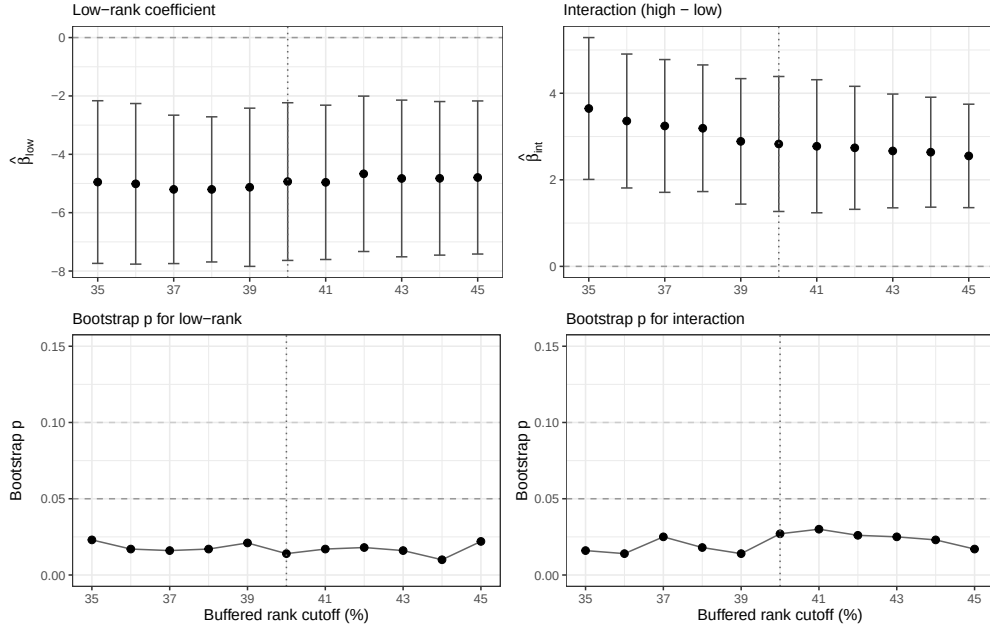


(c) Wellbeing and motivation

Figure K.6: Peer-exposure estimates in the full sample and among school-collapsed students.

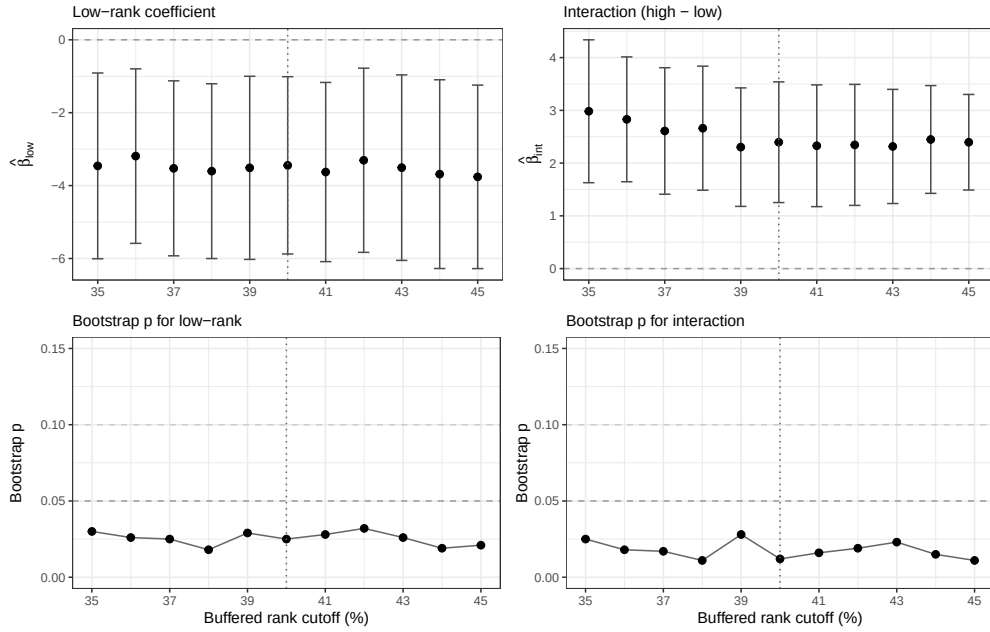
Notes. Filled circles and solid lines report peer-exposure estimates from the restricted sample containing only students whose own school collapsed. Hollow triangles and dotted lines report peer-exposure estimates from the full-sample joint specification. Hollow circles and dashed lines report the full-sample own-exposure estimates. Labels report wild cluster bootstrap p -values for the two peer-exposure estimates. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. All specifications include school-grade FE and an advanced-track indicator, age, gender, Tibetan minority, and class size. The academic specifications also include the relevant 2008 test score and a missing-score indicator. In the restricted sample, the specification omits own exposure because it is constant. The full-sample peer- and own-exposure estimates are jointly estimated. Higher values indicate better academic performance in Panel a, worse psychosocial conditions in Panel b, and better outcomes in Panel c.

Rank-cutoff sensitivity: Chinese score



(a) Chinese score

Rank-cutoff sensitivity: Total score



(b) Total score

Figure K.7: Sensitivity of peer-exposure rank heterogeneity to the buffered rank cutoff.

Notes. Each panel varies the symmetric school-grade rank cutoff used to define low- and high-rank groups from 35 to 45 percent. Upper subpanels report the low-rank coefficient on peer exposure ($\hat{\beta}_{low}$) and the interaction coefficient ($\hat{\beta}_{int}$), with whiskers giving 95% confidence intervals from classroom-clustered standard errors. Lower subpanels report the corresponding wild cluster bootstrap p -values (999 replications, impose-null Rademacher). Dotted vertical lines mark the 40/40 specification used in Table 2.

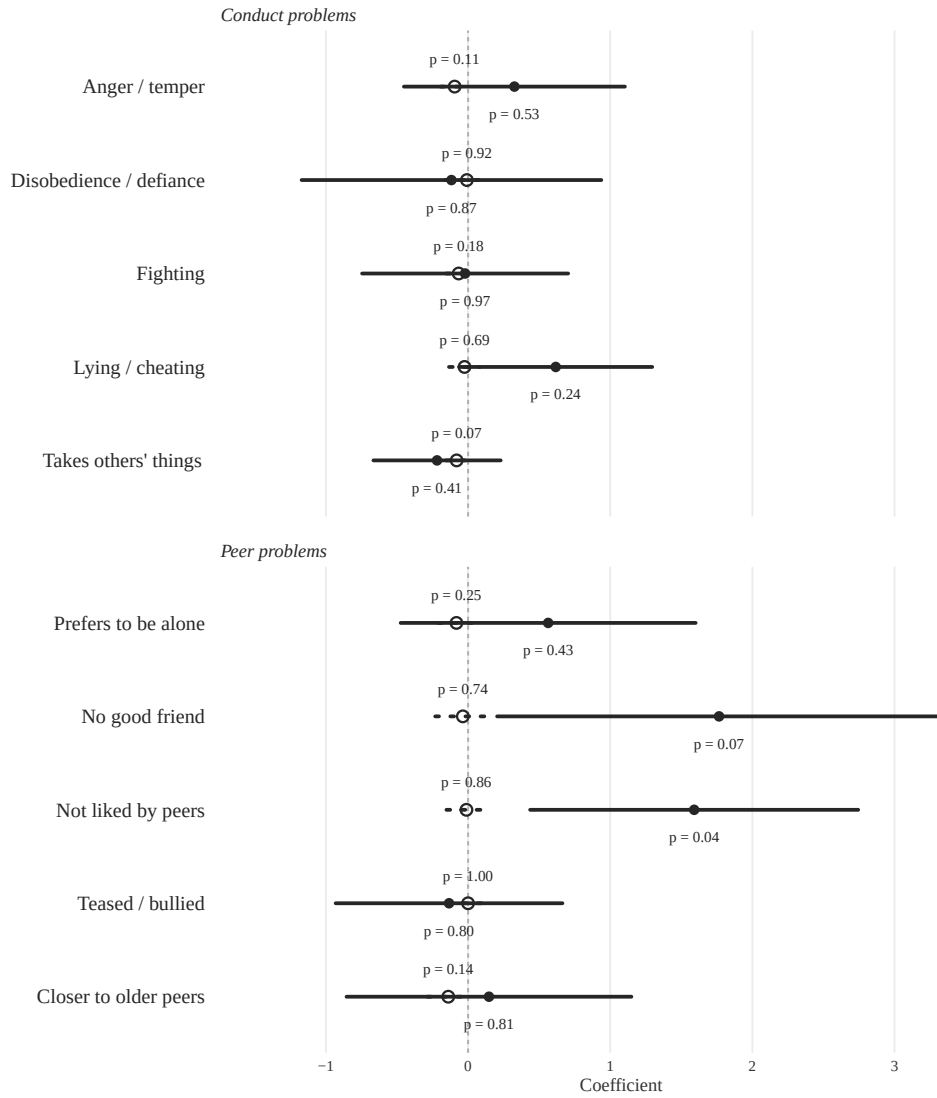


Figure K.8: Item-level SDQ outcomes, by subscale.

Notes. Items are grouped by SDQ subscale: conduct problems (top) and peer problems (bottom). Filled markers (solid line) report the effect of peer exposure, measured as the share of classmates whose old school collapsed. Hollow markers (dashed line) report the effect of own exposure. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. Marker labels report p-values based on wild cluster bootstrap inference where available and classroom-clustered p-values otherwise. Higher values indicate more behavioral or peer difficulties.

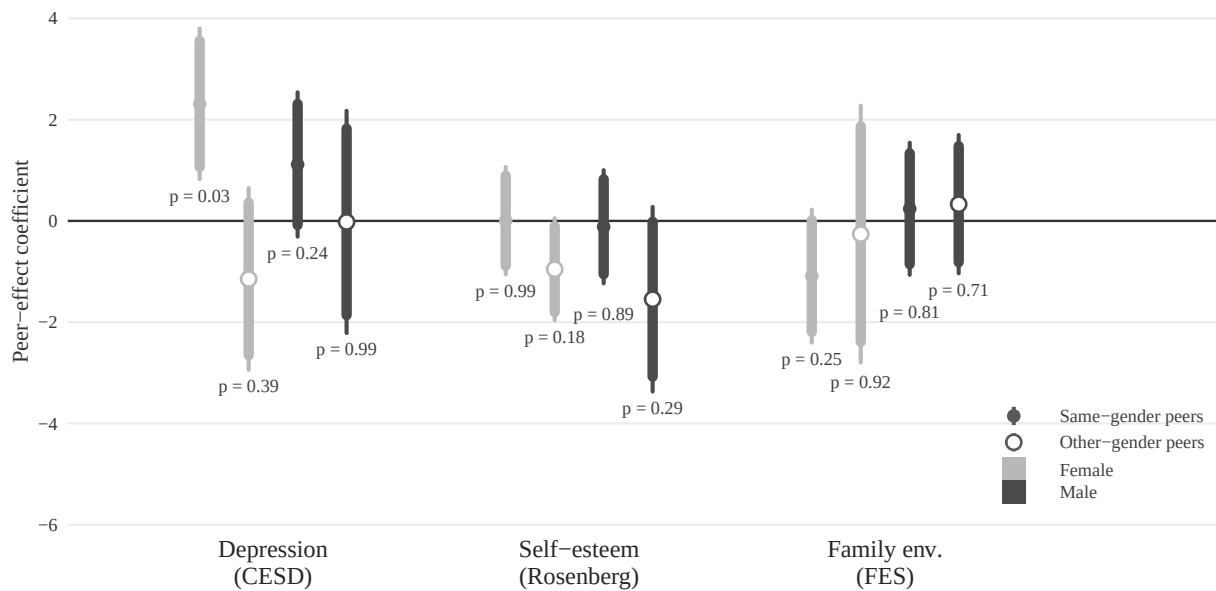


Figure K.9: Same- and other-gender peer effects on psychosocial outcomes, by recipient gender.

Notes. Each panel reports peer-exposure coefficients for three psychosocial outcomes: depression (CES-D), self-esteem (Rosenberg), and family environment (FES), all standardized so that higher values indicate worse outcomes. The sample is restricted to students whose own school collapsed. Same-gender peer exposure is the leave-one-out share of school-collapsed classmates of the same gender as the focal student; other-gender peer exposure is the corresponding share for classmates of the opposite gender. Coefficients are estimated separately for female and male students. Filled circles denote same-gender peer exposure; hollow circles denote other-gender peer exposure. Female estimates are shown in lighter gray and male estimates in darker gray. Wider intervals report 95% confidence intervals and narrower intervals report 90% confidence intervals, based on classroom-clustered standard errors. Labels report p -values from wild cluster bootstrap inference (999 replications, impose-null Rademacher). Controls are school-grade FE and an advanced-track indicator, age, Tibetan minority, and class size.

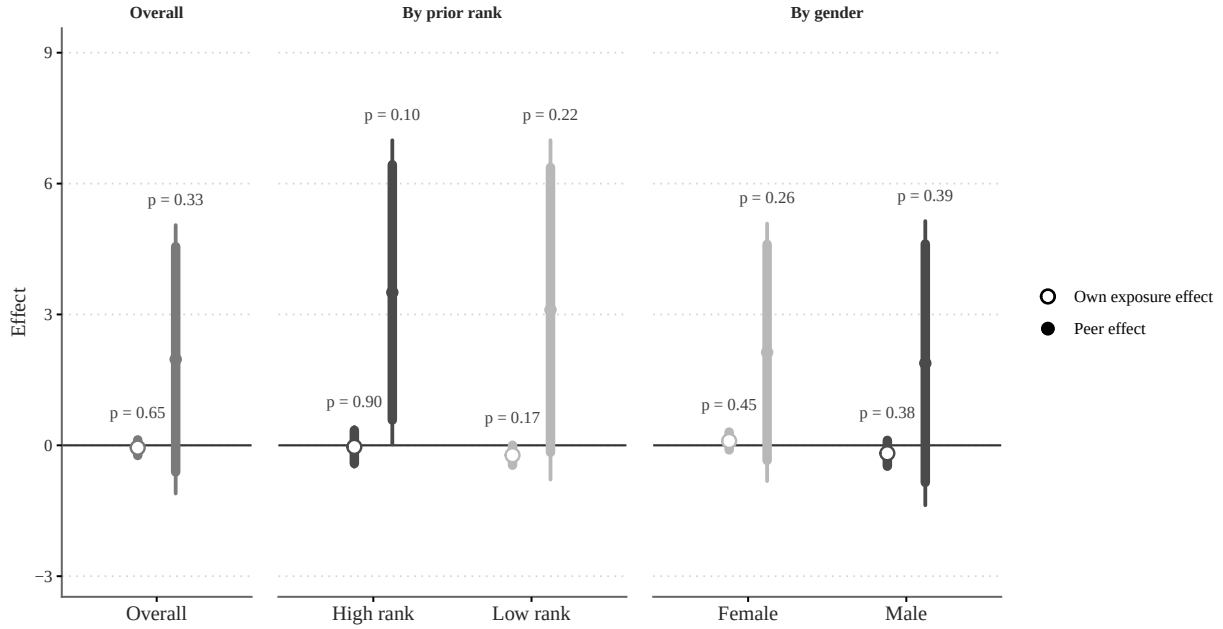
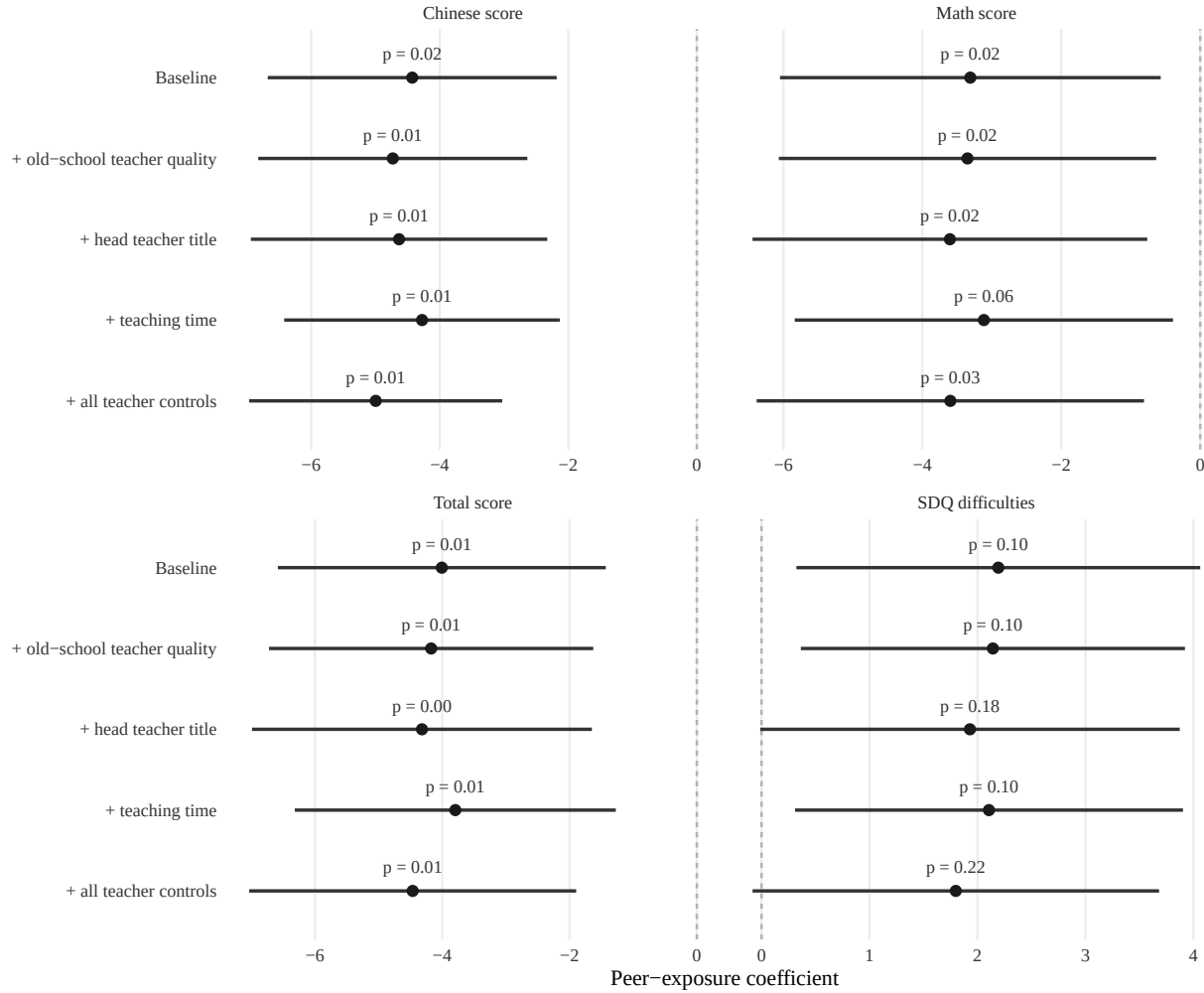


Figure K.10: Study-effort estimates: overall, by prior rank, and by gender.

Notes. The figure reports study-hours estimates for three comparisons: the full sample, the prior-rank split, and the gender split. Filled circles denote peer exposure, and hollow circles denote own exposure. In the prior-rank panel, lighter gray corresponds to low-rank students and darker gray to high-rank students. In the gender panel, lighter gray corresponds to female students and darker gray to male students. Wider intervals report 95% confidence intervals and narrower intervals report 90% confidence intervals, based on classroom-clustered standard errors. Labels report p -values from wild cluster bootstrap inference (999 replications, impose-null Rademacher). All specifications include school-grade FE and an advanced-track indicator, age, Tibetan minority, and class size. Own-exposure regressions additionally control for peer exposure, while peer-exposure regressions are estimated on the collapsed-school subsample. Prior rank is defined using the top and bottom 40 percent of the 2008 within-school-grade distribution.

Figure K.11: Peer-exposure estimates with alternative teacher-side controls.



Notes. Each point reports the peer-exposure coefficient from the classroom-exposure specification and from versions that additionally control for old-school teacher quality, the current head teacher's title, teaching time, and all three together. Chinese, math, and total score use the same student-level specifications as Table 1. SDQ difficulties uses the collapsed-school sample. Horizontal bars show 95 percent confidence intervals based on classroom-clustered standard errors.

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